11694 - Is PSR J2322-2650b really a carbon planet?

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

Folder	***Observation***	***Label***	***Observing Template***	***Science Target***
G140H				
	1		NIRSpec Fixed Slit Spectroscopy	(1) PSR-J2322-2650
G395H				
	2		NIRSpec Fixed Slit Spectroscopy	(1) PSR-J2322-2650
	3		NIRSpec Fixed Slit Spectroscopy	(1) PSR-J2322-2650

ABSTRACT

Molecular carbon (C2 and C3) has recently been detected in JWST spectra of the ``pulsar planet'' PSR J2322-2650b. These observations have been used to argue that the planet is a unique and exotic object: helium-dominated but highly enriched in carbon, with a C/O ratio exceeding 100. Such a

composition would severely challenge all known formation mechanisms. However, the inference of helium dominance is based only on the planet's low density (which is comparable to that of a typical hot Jupiter) as well as the lack of hydrogen-bearing molecules, and the C/O constraint assumes equilibrium chemistry, which may not be valid given the high-energy irradiation from the pulsar host.

We propose to test the helium-dominated hypothesis by using NIRSpec/G140H to look for escaping helium in the 1083 nm metastable line. We also propose to test the C/O constraint using NIRSpec / G395H to look for CO at wavelengths where it is least likely to be obscured by C2. Direct observational verification of these indirect inferences will bring us closer to understanding the true origin of this extreme and enigmatic object.

OBSERVING DESCRIPTION

With NIRSpec/G140H, we will observe the night side of the planet for two hours. With NIRSpec/G395H, we will observe the day side of the planet twice for two hours each.   Taking into account the 1 h scheduling uncertainty, we will have at least 30 min of observations on either side of conjunction for all visits.

For all visits, we opt to use the narrowest slit (0.2") to minimize the risk of contaminating sources in the background. The G140H observations should be nodded, with 5 nod positions and 2 integrations of 461 groups at each position. For the G395H observations, we use fewer groups per integration and forgo nodding because stability and time resolution are more important than avoiding a few bad pixels.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	PSR-J2322-2650	RA: 23 22 34.6384 (350.6443267d) Dec: -26 50 58.40 (-26.84956d) Equinox: J2000	Proper Motion RA: -2.26 mas/yr Proper Motion Dec: -8.31 mas/yr Parallax: 0.00159" Epoch of Position: 2022.4	
Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM. Category=Star Description=[Pulsars] Extended=NO				
(2)	acq_star	RA: 23 22 36.2857 (350.6511904d) Dec: -26 51 23.86 (-26.85663d) Equinox: J2000	Proper Motion RA: 6.5503 mas/yr Proper Motion Dec: -4.7632 mas/yr Parallax: 0.001885" Epoch of Position: 2016	
Comments: Category=Star Description=[K dwarfs] Extended=NO				

Observation

Proposal 11694, Observation 1

Wed May 20 17:00:32 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Observation 1) Warning (Form): The slew between the acquisition exposure and the farthest science exposure is 46.544 Arcsec (larger than the recommended limit of 38.000 Arcsec) and may result in reduced or no schedulability. See more information in the diagnostic browser.

(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	PSR-J2322-2650	RA: 23 22 34.6384 (350.6443267d) Dec: -26 50 58.40 (-26.84956d) Equinox: J2000	Proper Motion RA: -2.26 mas/yr Proper Motion Dec: -8.31 mas/yr Parallax: 0.00159" Epoch of Position: 2022.4	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Pulsars]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	2 acq_star	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	165648

Template

HFF Readout Mode	Slit	Subarray
false	S200A1	SUBS200A1

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	5	NONE

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G140H/F100LP	S200A1	NRSRAPID	461	2	1	NONE	5	10	7198.165	262575

Special Requirements	
	Phase 0.055 to 0.185 with period 0.322963997 Days and zero-phase 2456130.85411 HJD

Observation

Proposal 11694, Observation 2

Wed May 20 17:00:33 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Observation 2) Warning (Form): The slew between the acquisition exposure and the farthest science exposure is 45.718 Arcsec (larger than the recommended limit of 38.000 Arcsec) and may result in reduced or no schedulability. See more information in the diagnostic browser.

(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	PSR-J2322-2650	RA: 23 22 34.6384 (350.6443267d) Dec: -26 50 58.40 (-26.84956d) Equinox: J2000	Proper Motion RA: -2.26 mas/yr Proper Motion Dec: -8.31 mas/yr Parallax: 0.00159" Epoch of Position: 2022.4	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Pulsars]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	2 acq_star	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	165648

Template

HFF Readout Mode	Slit	Subarray
false	S200A1	SUBS200A1

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	NONE	NONE

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395H/F290LP	S200A1	NRSRAPID	306	16	1	NONE	1	16	7653.224	262575

Special Requirements	
	Phase 0.555 to 0.685 with period 0.322963997 Days and zero-phase 2456130.85411 HJD

Observation

Proposal 11694, Observation 3

Wed May 20 17:00:33 GMT 2026

Diagnostic Status: Warning

Observing Template: NIRSpec Fixed Slit Spectroscopy

Diagnostics

(Observation 3) Warning (Form): The slew between the acquisition exposure and the farthest science exposure is 45.718 Arcsec (larger than the recommended limit of 38.000 Arcsec) and may result in reduced or no schedulability. See more information in the diagnostic browser.

(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.

Fixed Targets

#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
(1)	PSR-J2322-2650	RA: 23 22 34.6384 (350.6443267d) Dec: -26 50 58.40 (-26.84956d) Equinox: J2000	Proper Motion RA: -2.26 mas/yr Proper Motion Dec: -8.31 mas/yr Parallax: 0.00159" Epoch of Position: 2022.4	

Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.

SIMBAD listed proper motion for this target. When retrieving targets with PM from SIMBAD, APT requests the coordinates be calculated with an epoch of the year 2000. Do not modify this epoch. Always review coordinates using the Target Confirmation tool, which graphically displays the PM.
Category=Star
Description=[Pulsars]
Extended=NO

Acquisition

#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID
1	2 acq_star	WATA	SUB32	CLEAR	NRSRAPID	3	1	1	0.08	165648

Template

HFF Readout Mode	Slit	Subarray
false	S200A1	SUBS200A1

Dithers

#	Primary Dither Positions	Sub-Pixel Pattern
1	NONE	NONE

Spectral Elements

#	Grating/Filter	Slit	Readout Pattern	Groups/Int	Integrations/Exp	#	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
1	G395H/F290LP	S200A1	NRSRAPID	306	15	1	NONE	1	15	7174.897	262575

Special Requirements	
	Phase 0.555 to 0.685 with period 0.322963997 Days and zero-phase 2456130.85411 HJD